

Definition Let $V : \text{IPS}/\mathbb{R}$, $\dim V = 3$. **Rotation** in $V :=$ orthogonal $T \in \text{End}(V)$, $\det T = 1$.

Theorem tells us they are either $1_{n \times n}$, $\begin{pmatrix} 1 & & \\ & -1 & \\ & & -1 \end{pmatrix}$ or $\begin{pmatrix} 1 & \\ & R(\theta) \end{pmatrix}$ under some suitable ONB, i.e.

$$\begin{pmatrix} 1 & \\ & R(\theta) \end{pmatrix}, \quad \theta \in \mathbb{R},$$

with no constraints on θ up to mod 2π . This rotation can be described as “take v_1 as axis and rotate by θ ”.

Let $V : \mathbb{R}$ -vs, $\dim = n$. Fix an ordered basis $e_1, \dots, e_n$ (may assume $V = \mathbb{R}^n$).

Recall $\{T \in \text{End}(V) \text{ invertible}\} \xleftrightarrow{1:1} \{\text{ordered basis in } V\}$.

In the context of geometry, we define **frames** in $V :=$ ordered basis of V . We say two frames have the same **orientation** if $\exists T \in \text{End}(V)$, $\det T > 0$ s.t. $(Tv_1, \dots, Tv_n) = (w_1, \dots, w_n)$. One can show this is an equivalence relation on frames.

Now, assume $V : \text{IPS}/\mathbb{R}$, $\dim V = n$. Fix $e_1, \dots, e_n : (\text{ordered}) \text{ ONB}$.

Orthogonal frames $:=$ frames $\underbrace{v_1 \dots v_n}_{\text{ONB}}$ of V . Hence

$$\{T \in \text{End}(V) \text{ orthogonal}\} \xleftrightarrow{1:1} \{\text{orthogonal frames of } V\},$$

$$\{T \in \text{End}(V) \text{ orthogonal}, \det = 1\} \xleftrightarrow{1:1} \{\text{orthogonal frames of } V \text{ with same orientation as } e_i\}$$

Specifying $n = 3$ gives

$$\{\text{rotations in } \mathbb{R}^3\} \xleftrightarrow{1:1} \{\text{positive oriented orthogonal frames in } \mathbb{R}^3\}.$$

Definition $\forall u \in \mathbb{R}^3, \theta \in \mathbb{R}$, define $R_u(\theta) :=$ “rotate around u by an angle of θ ”.

From above we know $\exists$ orthogonal frame u_1, u_2, u_3 , positive oriented s.t.

$$R_u(\theta) = \begin{pmatrix} 1 & & \\ & \cos \theta & -\sin \theta \\ & \sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} 1 & \\ & R(\theta) \end{pmatrix}$$

under ONB u_1, u_2, u_3 . Note that u_2, u_3 may differ by a rotation in $\langle u_1 \rangle^\perp$, and any valid selection of u_2, u_3 yield the same $R_u(\theta)$.

Examples

$$\begin{aligned}
 R_{e_1}(\theta) &= \begin{pmatrix} 1 & & \\ & \cos \theta & -\sin \theta \\ & \sin \theta & \cos \theta \end{pmatrix}, \\
 R_{e_2}(\theta) &= \begin{pmatrix} 1 & & \\ & \cos \theta & -\sin \theta \\ & \sin \theta & \cos \theta \end{pmatrix} \quad (\text{under ONB } e_2, e_3, e_1) \\
 &= \begin{pmatrix} \cos \theta & \sin \theta & \\ & 1 & \\ -\sin \theta & \cos \theta & \end{pmatrix}, \\
 R_{e_3}(\theta) &= \begin{pmatrix} \cos \theta & -\sin \theta & \\ \sin \theta & \cos \theta & \\ & & 1 \end{pmatrix}.
 \end{aligned}$$

Goal: express general rotations in terms of simpler ones (ideally R_{e_i}).

Assume $T \leftrightarrow$ positive oriented orthonormal frame (u_1, u_2, u_3) .

§ Euler's angles

Procedure: Fix T, u_1, u_2, u_3 as above. Note that all intermediate results are positive oriented orthonormal frames of $\mathbb{R}^3$.

- Set $f_2 := \begin{cases} e_3 \times u_3 & \text{if } e_3, u_3 \text{ are l.indep} \\ e_2 & \text{otherwise} \end{cases}$, then $f_2 \perp u_3, e_3$. Now take ψ s.t. $R_{e_3}(\psi)e_2 = f_2$.
Now $R_{e_3}(\psi) : (e_1, e_2, e_3) \mapsto (f_1, f_2, e_3)$ for some f_1 . (Note that $e_2 \perp e_3$.)
- $e_3, u_3 \perp f_2$, so may take θ s.t. $R_{f_2}(\theta)e_3 = u_3$.
Now $R_{f_2}(\theta) : (f_1, f_2, e_3) \mapsto (g_1, f_2, u_3)$ for some g_1 .
- $g_1, u_1 \perp u_3$, so may take φ s.t. $R_{u_3}(\varphi)g_1 = u_1$.
Now $R_{u_3}(\varphi) : (g_1, f_2, u_3) \mapsto (u_1, \blacksquare, u_3)$. Since this is a positive oriented orthonormal frame, we can conclude $\blacksquare = u_2$.

Theorem Let T be a rotation in $\mathbb{R}^3$ corresponding to (u_1, u_2, u_3) , then

$$T = R_{u_3}(\varphi)R_{f_2}(\theta)R_{e_3}(\psi).$$

We say T is expressed by the **Euler angles** (φ, θ, ψ) .

Proof Both sides carry (e_1, e_2, e_3) to (u_1, u_2, u_3) . □

Theorem We also have $T = R_{e_3}(\psi)R_{e_2}(\theta)R_{e_3}(\varphi)$.

Lemma Let P be an orthogonal transformation, $\varepsilon := \det P \in \{\pm 1\}$. Then $\forall \|u\| = 1, \theta \in \mathbb{R}$,

$$PR_u(\theta)P^{-1} = R_{Pu}(\varepsilon\theta).$$

Proof Take a positive oriented orthonormal frame $(u, v, w) \implies$ so is $(Pu, Pv, \varepsilon Pw)$.
To show: $PR_u(\theta) = R_{Pu}(\varepsilon\theta)P$.

$$\begin{aligned} R_{Pu}(\varepsilon\theta)P : u &\mapsto Pu \mapsto Pu, \\ v &\mapsto Pv \mapsto Pv \cos \varepsilon\theta + \varepsilon Pw \sin \varepsilon\theta, \\ \varepsilon w &\mapsto \varepsilon Pw \mapsto -Pv \sin \varepsilon\theta + \varepsilon Pw \cos \varepsilon\theta. \\ \\ PR_u(\theta) : u &\mapsto u \mapsto Pu, \\ v &\mapsto v \cos \theta + w \sin \theta \mapsto Pv \cos \theta + Pw \sin \theta, \\ w &\mapsto -v \sin \theta + w \cos \theta \mapsto -Pv \sin \theta + Pw \cos \theta. \end{aligned}$$

$\implies R_{Pu}(\theta)P, PR_u(\theta)$ have the same effect on (u, v, w) . □

Proof of Theorem Recall that $f_2 = R_{e_3}(\psi)e_2$.
Then $R_{f_2}(\theta) \stackrel{\text{Lem}}{=} R_{e_3}(\psi)R_{e_2}(\theta)R_{e_3}(\psi)^{-1}$, or

$$R_{f_2}(\theta)R_{e_3}(\psi) = R_{e_3}(\psi)R_{e_2}(\theta).$$

Also note that $u_3 = R_{f_2}(\theta)e_3 = R_{f_2}(\theta)R_{e_3}(\psi)e_3$, hence

$$R_{u_3}(\varphi) = R_{f_2}(\theta)R_{e_3}(\psi)R_{e_3}(\varphi)R_{e_3}(\psi)^{-1}R_{f_2}(\theta)^{-1},$$

which gives

$$\begin{aligned} R_{u_3}(\varphi)R_{f_2}(\theta)R_{e_3}(\psi) &= R_{f_2}(\theta)R_{e_3}(\psi)R_{e_3}(\varphi) \\ &= R_{e_3}(\psi)R_{e_2}(\theta)R_{e_3}(\psi). \end{aligned}$$

□

In terms of matrices,

$$T = \begin{pmatrix} \cos \psi & -\sin \psi & & \\ \sin \psi & \cos \psi & & \\ & & 1 & \\ & & & 1 \end{pmatrix} \cdot \begin{pmatrix} \cos \theta & \sin \theta & & \\ & 1 & & \\ -\sin \theta & \cos \theta & & \\ & & & 1 \end{pmatrix} \cdot \begin{pmatrix} \cos \varphi & -\sin \varphi & & \\ \sin \varphi & \cos \varphi & & \\ & & 1 & \\ & & & 1 \end{pmatrix}.$$

Remark $\exists$ other conventions. Different (φ, θ, ψ) may yield same rotations due to topological reasons. A simple example is $\theta = 0, \varphi + \psi = 0$ all returns id .

Next: another (better) way to describe rotations in $\mathbb{R}^3$.

§ Quaternions

Definition Let $\mathbb{H}$ be a $\mathbb{R}$ -vs with $\dim = 4$ and basis $1, i, j, k$.

- Addition and subtraction is immediately derived from the vector space structure.
- Multiplication $\mathbb{H} \times \mathbb{H} \rightarrow \mathbb{H}$ is $\mathbb{R}$ -bilinear, hence defining the multiplication of $\{1, i, j, k\}$ is enough.
 - $\forall x \in \{1, i, j, k\}, 1 \cdot x = 1 = x \cdot 1$,
 - $i^2 = j^2 = k^2 = -1$,
 - $ij = k = -ji, jk = i = -kj, ki = j = -ik$.

$\implies$ This defines $\mathbb{H} \times \mathbb{H} \rightarrow \mathbb{H}$. Can check: $(\mathbb{H}, +, 0, \cdot, 1)$ is a ring!

Distributivity is clear from bilinear. However $\mathbb{H}$ is not commutative from definition.

From vector space structure we can write

$$q = a + bi + cj + dk \in \mathbb{H}, \quad a, b, c, d \in \mathbb{R},$$

where a, b, c, d are unique. Note that $\mathbb{R}, \mathbb{C} \hookrightarrow \mathbb{H}$ are natural.